MIL-C-47200(MI)
12 July 1974
SUPERSEDING
MIS 15027
28 May 1965

MILITARY SPECIFICATION

COMPOUND, COATING, POLYURETHANE, ELECTRICAL INSULATING, ROOM TEMPERATURE CURE

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a room-temperature curing, polyurethane compound suitable for coating printed wiring and electronic assemblies.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-A-250	Aluminum Alloy 2024, Plate and Sheet 2024
TT-P-143	Paint, Varnish, Lacquer, and Related Materials, Packaging, Packing and Marking of

FSC 5970

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PPP-C-96	Can, Metal, 28 Gage and Lighter

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-F-5566	Fluid, Anti-Icing (Isopropyl Alcohol)
MIL-P-13949	Plastic Sheet, Laminated, Copper Clad (For Printed Wiring)

STANDARDS

Federal

FED-STD-141	Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 115	Testing Varnishes Used for Electrical Insulation

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

Uniform Classification Committee

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Materials. The compound shall consist of an isocyanate-modified, urethane polymer dissolved in an aromatic hydrocarbon or in a hydrocarbon-and-ester solvent mixture. The materials used shall be suitable for the purpose intended, and when processed, shall form a product conforming to all requirements of this specification.

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3.2 Product characteristics.

3.2.1 Flash point. The flash point of the compound shall be not less than 27 degrees Celsius (C) (81 degrees Fahrenheit (F)). (See Table II).

3.2.2 Solids content. The solids content of the compound shall be 47 plus or minus 6 percent by weight. (See Table II).

3.2.3 Viscosity. The viscosity of the compound shall be 100 to 450 centpoises. (See Table II).

3.2.4 Drying time. A surface coated with the compound shall be dry in 3 to 5 hours after application at 25 degrees plus or minus 5 degrees (77 degrees plus or minus 9 degrees F). (See 4.8.3).

3.2.5 Cure. The compound shall be cured sufficiently after drying for 24 hours at 25 degrees plus or minus 5 degrees C (77 degrees plus or minus 9 degrees F) and 50 plus or minus 20 percent relative humidity, to resist the solvent action of isopropyl alcohol. (See 4.8.4) If necessary, a non-foaming type catalyst supplied with the compound shall be used to effect cure.

3.2.6 Adhesion. The coating shall adhere tightly, without lifting or scaling to the surface of a coated panel. When cut as specified in 4.8.5, the coating shall not flake, chip or powder. The knife cut shall show beveled edges.

3.2.7 Transparency. The coating shall be transparent. (See 4.8.6).

3.2.8 Repairability. The coating shall be removable through sublimation and slight rubbing at 186 degrees plus or minus 5 degrees C (367 degrees plus or minus 9 degrees F) to permit repairs (See 4.8.7).

3.2.9 Insulation resistance. The insulation resistance shall have the value specified in Table I when measured under the corresponding environment. (See 4.8.8).

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Table I

Insulation resistance

Environment	Resistance, minimum
25 degrees plus or minus 5 degrees C (77 degrees plus or minus 9 degrees F)	400,000 megohms
140 degrees plus or minus 5 degrees C (284 degrees plus or minus 9 degrees F)	200 megohms
Method 103 of MIL-STD-202	100 megohms

3.2.10 Thermal shock. The coating shall exhibit no cracking, crazing or separation from the test panel, after exposure to Method 107, Test Condition B, of MIL-STD-202. (See Table II).

3.2.11 Fungus resistance. The coating shall not support fungus growth. (See Table II).

3.3 Workmanship. The compound shall be processed in such a way as to ensure compliance with the requirements of this specification. Freshly-opened containers of the compound shall be free from grit, sediment, and suspended matter. (See 4.8.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to

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perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections of the compound are classified as follows:

a. Quality conformance inspection, see 4.4.

b. Preparation for delivery inspection, see 4.5.

4.3 Inspection lot. For purposes of sampling, an inspection lot shall consist of a single batch of the compound offered for inspection at one time. Each batch shall have been produced in a single vat in no more than a 24-hour continuous production period with no change in process or raw materials.

4.4 Quality conformance inspection.

4.4.1 Quality conformance sampling and testing. The quality conformance inspection shall consist of the tests in Table II. A 1-quart sample of the compound shall be taken from the inspection lot and used in the tests in Group I of Table II. An additional quantity shall be taken from the lot to coat test panels as specified in 4.8.1. These panels shall be subjected to the tests in Group II of Table II.

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4.5 Preparation for delivery inspection. Each lot offered for acceptance shall be subjected to the preparation for delivery inspection.

4.5.1 Sampling and testing. Samples shall be taken in accordance with the applicable specifications and standards referenced in 5.1, 5.2, and 5.3. When no sampling plan is specified by those specifications and standards, samples shall be taken in accordance with MIL-STD-105, AQL 2.5. The samples shall be inspected as specified in 4.8.9.

4.5.2 Action in case of failure. Unless other criteria are defined in the applicable documents referenced in 5.1 through 5.3 inclusive, failure of one specimen in one or more respects shall be charged as a single defective. If the quantity of defectives exceeds that permitted, the lot represented by the sample shall be rejected. Defective specimens shall not be delivered on contract or purchase order until all deficiencies have been corrected.

4.6 Inspection equipment and facilites. Inspection equipment shall be of adequate quality and quantity to permit performance of the required examinations and tests, and shall comply with the equipment described in the applicable inspection methods of this specification and the documents referred to herein. Inspection facilities shall be adequate to permit performance of the required inspection.

4.7 Standard inspection conditions. Unless otherwise specified, all inspections shall be performed under conditions of room ambient pressure, temperature, and humidity.

4.8 Inspection methods. The requirements specified herein shall be verified in accordance with the test methods listed in Table II, with modifications as specified in 4.8.2 through 4.8.8.

4.8.1 Preparation of test panels.

4.8.1.1 Type A. Type A test panels shall be prepared in accordance with the following procedure:

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Table II

Quality conformance inspection

Property	Panel type & quantity A				Panel type & quantity B				Test Method	Modified by	Req't
Group I											
Visual examination									Visual	4.8.2	3.3
Flash point									4291 of FED-STD-141	----	3.2.1
Solids content									4041 of FED-STD-141	----	3.2.2
Viscosity									ASTM: D115-55	----	3.2.3
Group II	3	3	3	3	3	3	3	3			
Drying time	x								4061 of FED-STD-141	4.8.3	3.2.4
Cure	x								----	4.8.4	3.2.5
Adhesion		x							----	4.8.5	3.2.6
Transparency					x				Visual	4.8.6	3.2.7
Repairability					x				----	4.8.7	3.2.8
Insulation resistance						x	x	x	----	4.8.8	3.2.9
Thermal shock			x						107, test condition B -- of MIL-STD-202		3.2.10
Fungus resistance				x					MIL-E-5272	----	3.2.11

x Indicates tests to be performed on a given set of 3 panels.

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a. Cut twelve panels from aluminum sheet conforming to QQ-A-355, to the following minimum dimensions: 0.064 by 2 by 4 inches.

b. Coat the panels to a thickness of 0.006 plus or minus 0.004 inch and allow them to air dry.

c. Not more than 5 hours after the coating operations has commenced, select three panels at random and subject them to the drying time test specified in 4.8.3.

d. Set aside all the panels to dry at 25 degrees plus or minus 5 degrees C (77 degrees plus or minus 9 degrees F) and 50 plus or minus 20 percent relative humidity for a minimum of 24 hours prior to use in the tests specified in Table II.

4.8.1.2 Type B panels. Type B test panels shall be prepared in accordance with the following procedure:

a. Etch a pattern in accordance with Figure 1 on Type FL-GB062C2 plastic sheet conforming to MIL-P-13949.

b. Brighten the clad surface by thoroughly scribbing the sheet, and allow it to air dry for a minimum of 24 hours.

c. Cut twelve 2 by 4-inch panels from the sheet.

d. Coat the copper-clad surface of the panels to a thickness of 0.006 plus or minus 0.004 inch and allow them to dry at 25 degrees plus or minus 5 degrees C (77 degrees plus or minus 9 degrees F) and 50 plus or minus 20 percent relative humidity for a minimum of 24 hours prior to use in the tests specified in Table II. Tests shall be completed within a period of less than 9 days.

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4.8.2 Visual examination. The compound in freshly-opened containers shall be visually examined to determine compliance with the requirements specified in 3.3. If necessary, small quantities of the compound shall be poured onto clean glass surfaces for examination.

4.8.3 Drying time test. Not more than 5 hours after application of the coating to the test panels (see 4.8.1), the drying time shall be verified in accordance with finger method for the tack-free stage of drying as specified in Method 4061 of FED-STD-141. (See 3.2.4).

4.8.4 Cure test. Test panels cured for a maximin of 24 hours from the time of application at 25 degrees plus or minus 5 degrees C (77 degrees plus or minus 9 degrees F) and 50 plus or minus 20 percent relative humidity, shall be immersed for at least 60 seconds in isopropyl alcohol conforming to MIL-F-5566. The panels shall be moved from the alcohol and tested immediately in accordance with the finger method for the tack-free stage of drying as specified in Method 4061 of FED-STD-141. (See 3.2.5).

4.8.5 Adhesive test. A narrow ribbon of the coating shall be cut from the test panel with a sharp knife blade, while holding the blade at an angle of approximately 30 degrees to the panel. The cut film shall be examined for compliance with the requirements in 3.2.6.

4.8.6 Transparency test. The test panels shall be examined to verify that the copper etching is visible through the coating. (See 3.2.7).

4.8.7 Repairability test. Using standard production techniques, the coating shall be removed from the test panel by sublimation at 186 degrees plus or minus 5 degrees C (366 degrees plus or minus 9 degrees F). (See 3.2.8.)

4.8.8 Insulation resistance tests. (See 3.2.9) The insulation resistance between the separate segments of foil on the face of the panels shall be measured using a 500-volt direct current (dc) megohmmeter on test panels prepared as specified in 4.8.1.2 and conditioned as follows:

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a. 3 panels cured at 25 degrees plus or minus 5 degrees C (77 degrees plus or minus 9 degrees F) for a minimum of 24 hours.

b. 3 panels cured at 25 degrees plus or minus 5 degrees C for a minimum of 24 hours, and then conditioned in a temperature chamber at 140 degrees plus or minus 5 degrees C (284 degrees plus or minus 9 degrees F) for a minimum of 15 minutes.

c. 3 panels cured at 25 degrees plus or minus 5 degrees C for a minimum of 24 hours, and then exposed to Method 103, Test Condition B, of MIL-STD-202.

4.8.9 Preparation-for-delivery inspection method. Inspection shall be performed to determine whether the compound is prepared for delivery in accordance with the requirements in 5.1, 5.2, and 5.3. (See 4.5).

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Level A. The compound shall be furnished in metal containers conforming to PPP-C-96. The size of the containers shall be as specified in the contract or purchase order. Containers shall be provided with wire handles which have been galvanized or protectively coated to resist corrosion.

5.1.2 Level C. Packaging shall be done in accordance with the supplier's commercial practice, and shall provide adequate protection against deterioration during shipment from the supply source to the first receiving activity.

5.2 Packing.

5.2.1 Level A. The containers shall be packed for overseas shipment in compliance with TT-P-143, Level A.

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5.2.2 Level B. The containers shall be packed in compliance with the domestic shipping requirements of TT-P-143, Level B.

5.2.3 Level C. The containers shall be packed for shipment in compliance with Uniform Freight Classification Ratings, Rules and Regulations or other carrier regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Identification. Containers of the compound shall be marked in compliance with MIL-STD-130. The following information shall appear on each container.

a. Name of the contents:

COATING COMPOUND, POLYURETHANE, ELECTRICAL INSULATING, ROOM TEMPERATURE CURE

b. This specification number.

c. Manufacturer's name or trademark.

d. Manufacturer's formula number.

e. Batch number.

f. Date of manufacturer.

g. Quantity.

WARNING: AVOID INHALATION OF FUMES OR EXCESSIVE SKIN CONTACT.

KEEP AWAY FROM FACE AND EYES.

WORK IN A WELL-VENTILATED AREA.

FLAMMABLE - KEEP AWAY FROM SPARKS AND OPEN FLAME.

5.3.2 Marking for shipment. Unit containers and exterior shipping containers shall be marked in compliance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The compound covered by this specification is intended for use as a moisture-resistant, insulating coating for electronic assemblies.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and packing required.

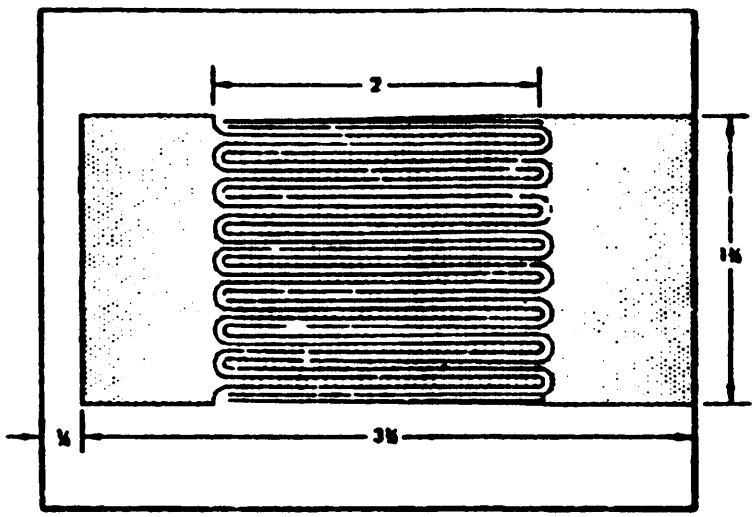
c. Size of container.

6.3 Supersession data. This specification includes the requirements of Missile Interim Specification, MIS-15027, dated 28 May 1965.

Custodian:
Army-MI

Preparing Activity:
Army-MI
Project No. 5970-A445

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NOTES:

1. Dimensions in inches.
2. Fingers

   Types: Standard (solid and hollow), Narrow, and Pointed
   Distance between adjacent fingers: 0.025.

3. Standard Fingers:

   Shape of Webs and Tips: Elliptical

   Tips:

   Major Diameter: 0.150
   Minor Diameter: 0.075

   Webs:

   Major Diameter: 0.250
   Minor Diameter: 0.125

4. Location of hollow (hollow fingers): Centered
5. Width of hollow: 0.025
6. Length of Hollow: 1.8
7. Width of Narrow Fingers: 0.025
8. Taper of pointed fingers: 2 degrees

Figure 1. Test specimen

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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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